

Physics 216: Electronics

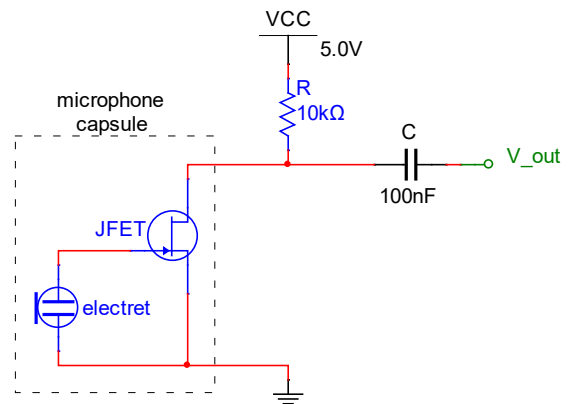
“M” problems

Part E: Sensors, Some Special Purpose Chips, and Applications

Answers to problems are at the end of this document.

ME1. When you used an electret microphone in the lab, you connected the microphone in series with a 10 kΩ resistor as shown below. Suppose that in the circuit shown, the voltage at the output of the microphone capsule has a DC level of 3 volts and an AC level of 40 mV peak-to-peak.

(a) Making the gross over-simplification that the JFET inside the microphone is behaving like a variable resistor, determine the JFET's range of effective resistance in the circuit. (b) Assuming the JFET's effective range of resistance is unchanged from the previous part, what would be the AC output level if you used an external resistance of $R = 1 \text{ k}\Omega$? (c) What would be the AC output level if you used an external resistance of $R = 100 \text{ k}\Omega$? (d) Write a rule of thumb for how to choose the value of external resistance to use in a JFET circuit like this one to maximize the AC output.

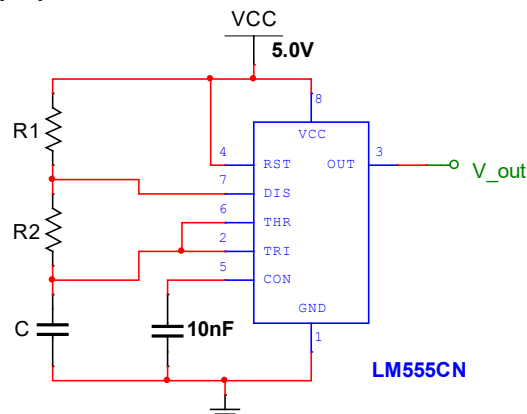


ME2. Derive the equations below for the dwell times of the 555 timer in astable mode:

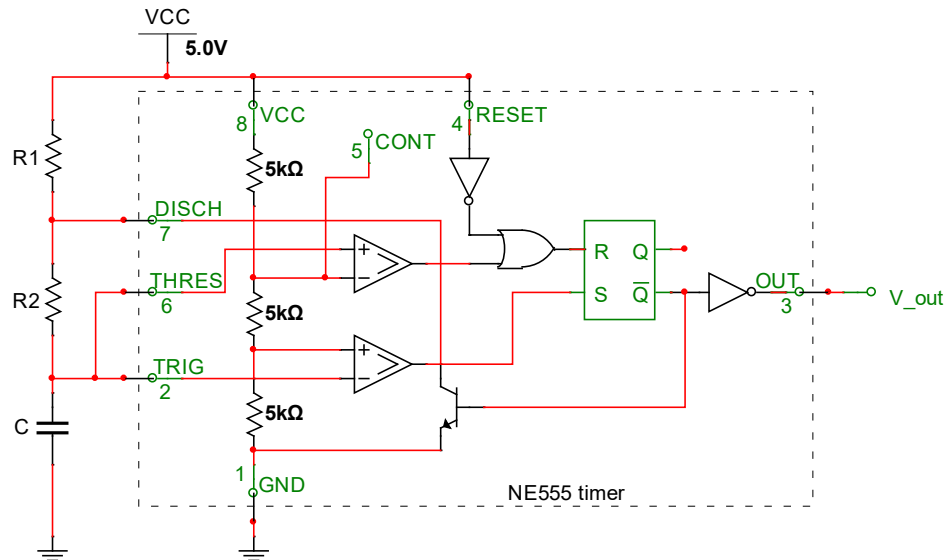
$$\Delta t_{HIGH} = \ln(2)C(R_1 + R_2)$$

$$\Delta t_{LOW} = \ln(2)CR_2$$

ME3. In the figure below, the 555 timer is wired in the astable mode, with $R_1 = 27 \text{ k}\Omega$, $R_2 = 10 \text{ k}\Omega$, and $C = 33 \text{ nF}$. (For this problem, it's fine to use the result of problem ME2, above. Those two equations were also given to you in part 3 of lab 23.) (a) Find the frequency of the oscillator. (b) Find the duty cycle of the oscillator.



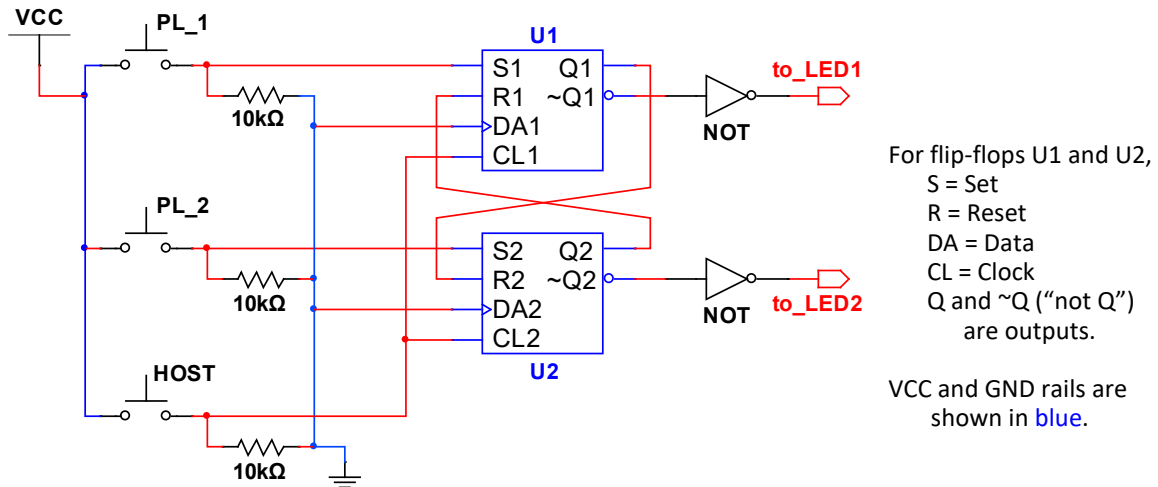
ME4. The circuit below shows the internal block diagram of a 555 timer, connected in the astable mode to a capacitor and two resistors. In this problem, you can assume that all digital voltages (R and S inputs of the flip-flop, $\bar{Q}$, and V_{out}) are all either 0 V or 5 V, even though they might differ slightly from those values. (a) What are the voltages at the V_- input of the upper comparator and the V_+ input of the lower comparator? (b) Complete the table below.



	While capacitor charging	At instant of max V_C	While capacitor discharging	At instant of min V_C
V_C (capacitor)	$1.6 \text{ V} < V_C < 3.3 \text{ V}$	$V_C > 3.3 \text{ V}$	$1.6 \text{ V} < V_C < 3.3 \text{ V}$	$V_C < 1.6 \text{ V}$
V_{out}	5 V		0 V	
Q				
$\bar{Q}$				
Transistor operating mode (cutoff, active, or saturation)				
$V_{disch} = 0$ or $I_{disch} = 0$				
flip-flop input R				
flip-flop input S				

ME5. The circuit below is for a quiz show in which two contestants each try to be the first to hit their button (or “buzz in”) and answer each question.

- After a question is read, each player hits their button (PL_1 or PL_2) as fast as they can.
- A light (LED1 or LED2) goes on over the contestant that buzzes in first.
- Only the fastest player’s light goes on; after that, the slower player’s button is “locked out” and has no effect.
- Before the next question, the host button is pressed to reset both player’s lights to “off.”



Initially, all switches are open and both lights are off, so $Q_1 = Q_2 = 0$. (We’ll use “1” and “0” here instead of specific voltages.)

- What is the purpose of the three 10 kΩ resistors to ground? (Hint: See Lab 8, Part 2.)
- If player 1 hits the button first, what inputs and outputs on U1 and U2 are changed while the switch PL_1 is closed? (Hint: of the twelve leads shown, four change values.)
- What changes when switch PL_1 is released (opened)?
- Player 2 hits their button later than player 1. What, specifically, happens to both Q_2 and to $\overline{Q_2}$ while button PL_2 is pressed? Why doesn’t Player 2’s light go on?
- It would seem that the circuit could be simplified by removing the two NOT gates and connecting the two lights directly to the Q_1 and Q_2 outputs. Why is that not quite the same as what is done above, using the NOT gates? (Hint: See Lab 22, Part 3, as well as the helpful figure in the answers here.)
- The host presses the host button to reset the players’ lights. Explain how this action affects both U1 and U2. (Hint: U1 and U2 behave differently here, due to the different values of their inputs.)

Answers:

ME1. (a) 14.752 kΩ to 15.253 kΩ (b) 9.8 mV peak-to-peak (c) 18.9 mV peak-to-peak.

ME3. (a) 930 Hz (b) 78.7%.

ME4. (a) 3.33 V and 1.67 V (b) Hints: Solve column 1 in this order: $\bar{Q}$, operating mode, I_{DISCH} , R , S . Solve columns 2, 3, and 4 in this order: R , S , $\bar{Q}$, operating mode, I_{DISCH} or V_{DISCH} , V_{OUT} .

ME5. (e) The figure below shows how a flip-flop's Q and $\bar{Q}$ depend on S and R .

